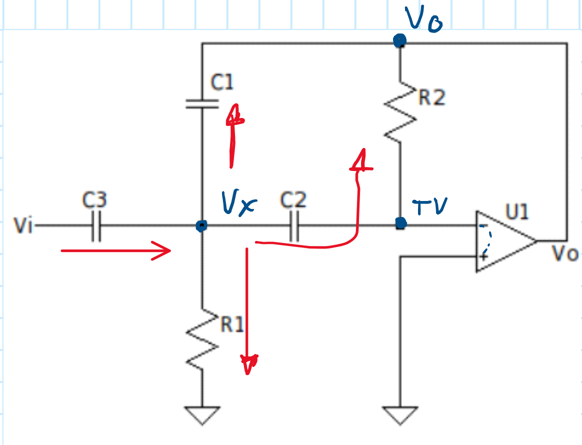
MFB



$$* \ (V_x - 0)\, sC_2 = (0 - V_o)\, G_2$$

$$\hookrightarrow V_x\, sC_2 = -V_o G_2 \rightarrow \boxed{V_x = -V_o \frac{G_2}{sC_2}}$$

$$* \ (V_i - V_x)\, sC_3 = V_x G_1 + (V_x - V_o)\, sC_1 + (-V_o G_2)$$

$$\rightarrow V_i\, sC_3 = V_x G_1 + V_x sC_1 - V_o sC_1 - V_o G_2 + V_x sC_3$$

$$\rightarrow V_i\, sC_3 = V_x (G_1 + sC_1 + sC_3) - V_o (sC_1 + G_2)$$

$$\rightarrow V_i\, sC_3 = -V_o \frac{G_2}{sC_2} (G_1 + sC_1 + sC_3 - V_o (sC_1 + G_2)$$

$$\rightarrow V_i\, sC_3 = V_o \left[\frac{-G_2 G_1 - sC_1G_2 - sC_3G_2 - s^2 C_1C_2 - sC_2G_2}{sC_2} \right]$$

$$\rightarrow V_i = V_o \left[\frac{-s^2 C_1 C_2 - s(C_2G_2 + C_3G_2 + C_1G_2) - G_1G_2}{s^2 C_2 C_3} \right]$$

$$\rightarrow H(s) = \frac{V_o}{V_i} = - \frac{s^2 C_2 C_3}{s^2 C_1 C_2 + s(C_2G_2 + C_3G_2 + C_1G_2) + G_1G_2}$$

$$\rightarrow H(s) = - \frac{s^2 \frac{C_3}{C_1}}{s^2 + s\left(\frac{G_2}{C_1} + \frac{C_3G_2}{C_1C_2} + \frac{G_2}{C_2}\right) + \frac{G_1G_2}{C_1C_2}}$$

$$\boxed{\rightarrow H(s) = - \frac{s^2 \frac{C_3}{C_1}}{s^2 + s\left(\frac{1}{R_2C_1} + \frac{C_3}{R_2C_1C_2} + \frac{1}{R_2C_2}\right) + \frac{1}{R_1R_2C_1C_2}}}$$